

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

ANSWER SCHEME

CLASS

INDEX NUMBER

BIOLOGY

9744/02

Paper 2 Structured Questions

29 August 2017

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	
8 OR 9	
Total	100

This document consists of **XX** printed pages.



Section A

Answer **all** questions.

- 1 Fig. 1.1 shows a ligand binding to the G Protein-coupled receptor (GPCR) which is embedded on the cell surface membrane.

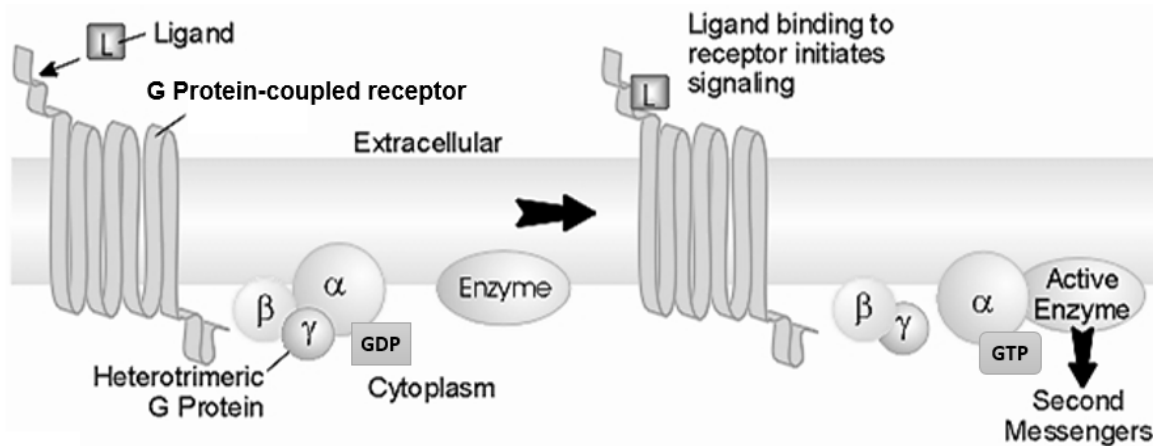


Fig. 1.1

- (a) (i) Identify the ligand in Fig. 1.1.

glucagon;;

[1]

- (ii) Explain why ligand mentioned in (a)(i) unable to pass through the cell surface membrane.

1. glucagon is a peptide hormone/ prot

which is a large macromolecule/ contains amino acids with charged R groups;;

2. it is unable to pass thru small temporary gaps in CSM/ will be repelled by the hydrophobic core;;
/ no tpt prot (carrier/ channel) large enough to facilitate its tpt across CSM;;

[2]

- (b) With reference to Fig 1.1, describe what happens when the ligand binds to the GPCR.

1. when ligand binds to GPCR, receptor is activated & undergoes conformational Δ

activated receptor then interacts with G prot bound with GDP;;

2. causing it to exchange its bound GDP with GTP

activating G prot, causing β & γ subunits of G prot to dissociate from α subunit;;

3. α subunit together with bound GTP then binds to enz to activate it

activated enz will then activate 2nd messengers involved in signal transduction;;

[3]

(c) With a named example, define "second messengers".

1. 2nd messengers are small, non-prot, water soluble mols

that relays signals from cell surface receptors to target mols in cell;;

2. e.g. cyclic AMP (adenosine monophosphate)/ Ca^{2+} / cyclic GMP (guanosine monophosphate)/ diacylglycerol;;

[2]

(d) Explain how intracellular signal is terminated when ligand is released from the receptor.

1. GTPase in α subunit of G prot will hydrolyse bound GTP to GDP;;

2. α subunit will associate together with β & γ subunits of G prot which forms inactive G prot;;

3. cAMP will be converted back to AMP by phosphodiesterase;;

[2]

(e) One of the side effects of a particular drug includes non-responsiveness of GPCR to ligands.

Suggest how the drug could have caused such non-responsiveness of GPCR to ligands.

1. (competitive inhibitor) drug has a similar 3D config as ligand (glucagon)

competes with ligand for ligand binding site ® active site

2. drug binds permanently to GPCR binding site / via strong permanent bonds

therefore, ligands unable to bind to GPCR to initiate signal transduction;;

[2]

@ non-competitive inhibition → binds to al__ site → Δ 3D config of GPCR → I__ b__ s__ ⊗ compl to ligand

[Total: 12]

2 Fig. 2.1 is a photomicrograph of plants cells, with some undergoing mitosis.

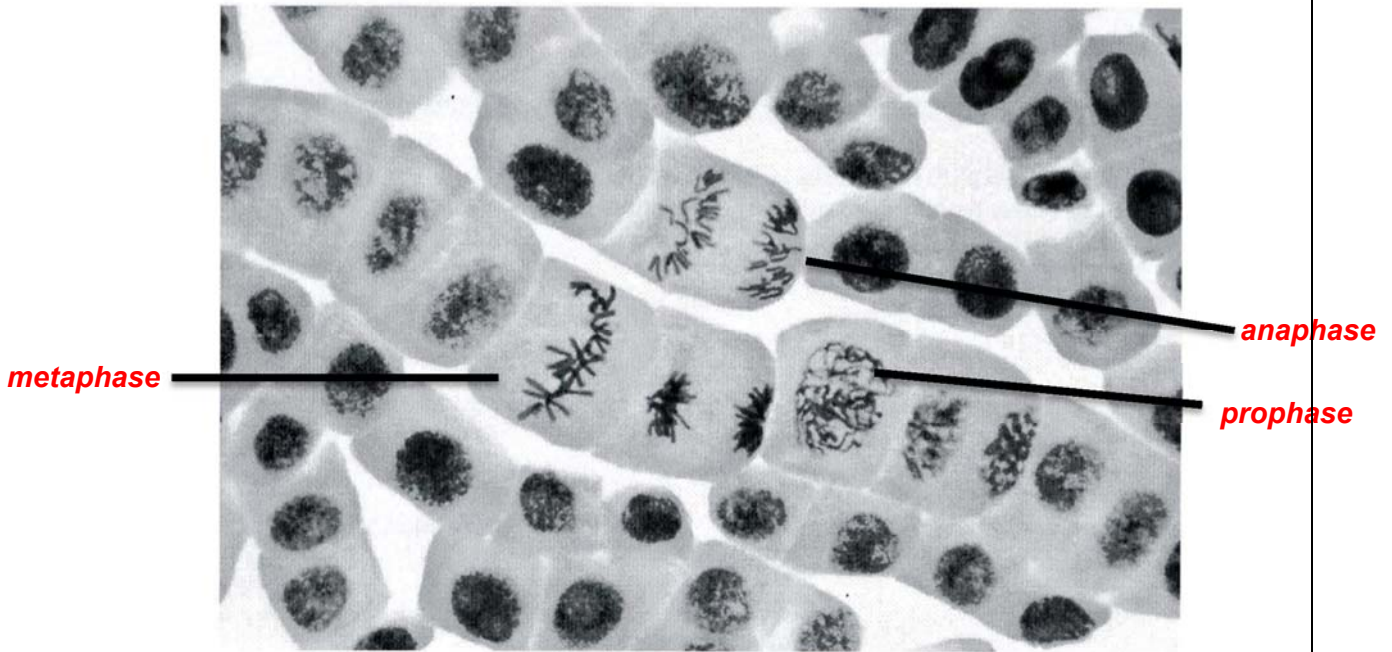


Fig. 2.1

- (a) On Fig 2.1, use labels and label lines to indicate **one** cell in anaphase stage of mitosis. [1]
- (b) The longest stage of the mitotic cell cycle, interphase, is divided into three phases, **G1, S** and **G2**.
- (i) Describe what happens in the G1 phase.
1. ***synthesis of organelles e.g. ER, mito etc***
 2. ***synthesis of prots (needed for S phase – replication of DNA);;***
 3. ***increase in cytoplasmic vol resulting in an increase in cell size;;***
 4. ***nucleolus actively synthesises ribosomal RNA for formation of ribosomes;;*** [3]
- (ii) There are various checkpoints in the mitotic cell cycle. One of them is present in the G1 phase called the G1 checkpoint.

Describe the function of G1 checkpoint.

1. ***checks for presence of growth factors;;***
2. ***checks for DNA damage;;***
3. ***checks for appropriate cell size and sufficient nutrients;;*** [2]

- (iii) When cell cycle checkpoints are defective, cancer could arise.

With reference to two named genes, outline the development of cancer.

1. gain-of-function mutation of proto-oncogene to oncogene (Ras gene)

loss-of-function mutation of normal TSG to mutated TSG (p53 gene);;

2. cells with mutations are able to evade apoptosis resulting in proliferations of cells with mutations

telomerase genes are activated in cells which prevents shortening of telomeres;;

3. resulting in cells having limitless replicative potential

leading to uncontrolled cell division & over-proliferation resulting in formation of a mass of overlapping cells – tumour;;

4. as tumour grows in size, activation of genes involved in angiogenesis results in proliferation of blood vessels to tumour cells

activation of genes involved in invasion & metastasis allows tumour to migrate to distant sites;;

[4]

- (b) Stem cells go through mitosis as well. But they go through asymmetrical division, where the fate of the two daughter cells are different.

- (i) State the potency level of adult stem cells and their function in our body.

1. multipotent;;

2. serve to maintain steady-state functioning of cells

**by generating replacements for cells lost through disease/
tissue injury;; @cell repair**

[2]

- (ii) Suggest one similarity between stem cells and cancer cells.

1. both have active telomerase gene to pdc active telomerase @ active telomerase gene only

to prevent shortening of telomeres;;

2. thereby enabling both stem cells & cancer cells to divide indefinitely;;

[2]

[Total: 14]

- 3 Pure breeding sweet pea plants with purple flowers and long pollen grains were crossed with pure breeding plants with red flowers and round pollen. All the F_1 plants had purple flowers and long pollen grains. These F_1 plants were then allowed to self-pollinate and the seeds produced were grown. The following results were obtained in this F_2 generation.

4831 purple flowers and long pollen grains
 390 purple flowers and round pollen grains
 393 red flowers and long pollen grains
 1338 red flowers and round pollen grains

- (a) Explain what is meant by the term "pure breeding".
homozygous for all genes (involved)/ having identical alleles for all genes (involved);; [1]

- (b) State the expected phenotypic ratio of the F_2 generation.
9 purple flowers & long pollen grains : 3 purple flowers & round pollen grains
3 red flowers & long pollen grains : 1 red flowers & round pollen grains [1]

- (c) Using suitable symbols, draw a genetic diagram to explain the observed results of the F_2 generation.

Legend:

Let F represent the dominant allele for purple flowers.

Let f represent the recessive allele for red flowers.

Let L represent the dominant allele for long pollen grains.

Let l represent the recessive allele for round pollen grains.

F_1 Phenotype: **Purple flower, long pollen grains** $\times$ **Purple flower, long pollen grains**

F_1 Genotype: **$FfLl$** $\times$ **$FfLl$** ;;

Crossing over during meiosis ;;

After meiosis, F_1 Gametes:

By random
fertilization,
Punnett
Square:

	FL	Fl	fL	fl
FL	FFLL	FFLl	FfLL	FfLl
Fl	FFLl	FFll	FfLl	Ffll
fL	FfLL	FfLl	ffLL	ffLl
fl	FfLl	Ffll	ffLl	ffll

;;

F_2

Genotype:

$F_L_$

F_ll

$ffL_$

$ffll$

;;

F_2

Phenotype:

Purple
flower, long
pollen

Purple
flower,
round
pollen

Red flower,
long pollen

Red flower,
round
pollen

Expected
Phenotypic
Ratio:

9:

3:

3:

1

;;

Observed
phenotype
numbers

4831

390

393

1338

Observed
phenotypic
ratio

12:

1:

1:

3

[5]

- (b) Suggest how similar crossing experiments with many different pairs of characters could be used to map the position of genes on the chromosomes of sweet pea plants.

1. if expected phenotypic ratio of 9:3:3:1 is observed

there is no linkage b/w two genes for that pair of characters are found on diff chr;;

2. if not, recombination freq which reflects dist of 2 genes on chr can be calculated;

using formula: no. of org showing recombinant phenotypes/ total no. of offspring x 100%

3. further apart two genes, higher possibility of crossing over;

higher recombination frequency; @vice versa

[3]

[Total: 10]

- 4 Fig. 4.1 shows the life-cycle of *Aedes aegypti*, which are often vectors of viral diseases like dengue fever, chikungunya and yellow fever.

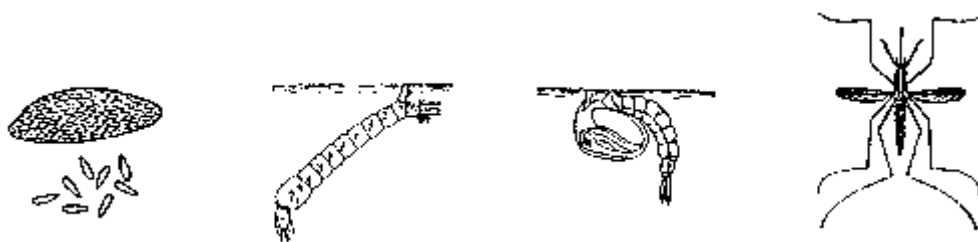


Fig. 4.1

- (a) With reference to Fig 4.1, name the four stages in a *Aedes aegypti* life-cycle.

eggs, larva, pupa, adult

[2]

- (b) The following is an extract from an article "*Record 2,441 dengue cases reported in Singapore for January*" published on Singapore's The Straits Time website on 2nd Feb 2016

"SINGAPORE - A total of 636 dengue cases were reported for the week of Jan 24 to 30 - the same number as the previous week - according to the latest figures released by the National Environment Agency (NEA) on Tuesday (Feb 2).

This brings the total number of cases for the first four weeks of the year to 2,441, an unusually high number for January given that it is traditionally the low season for dengue."

Fig 4.2 shows the weekly number of dengue cases in Singapore from 2013 to 2016

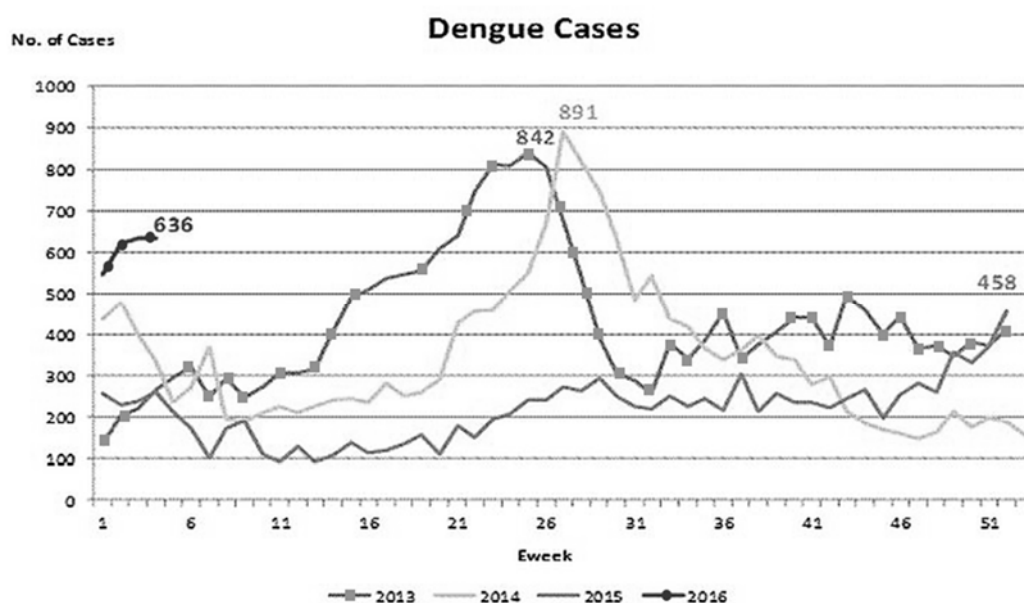


Fig. 4.2

- (i) With reference to Fig 4.2, describe the general trend in dengue cases in 2013.
1. ***in 2013, no. of dengue cases in SG shown a general increase in first half of the year;***
then it started to decrease & fluctuated in second half of the year;;
 2. ***from about 140 cases in beginning of year, it then increased to a peak of 842 cases in Week 25;***
then it decreased to about 410 cases at end of the year;
- [2]

Fig 4.3 shows average monthly temperature in Singapore in Year 2013.

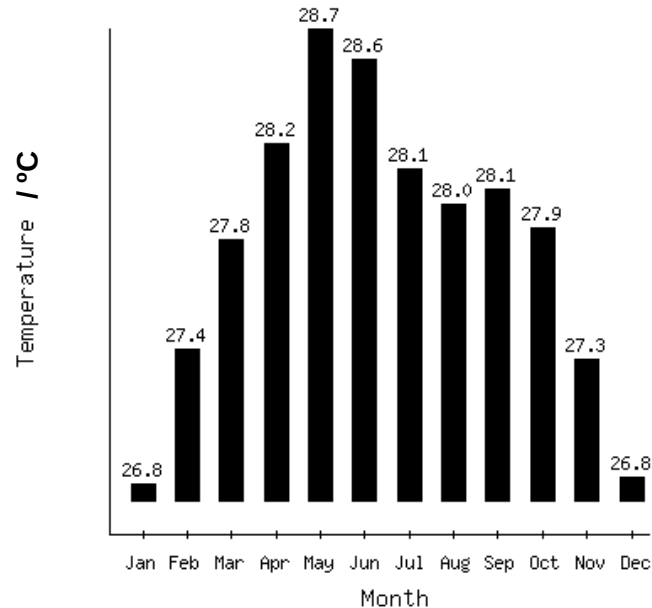


Fig 4.3

- (ii) With reference to Fig 4.3, describe the temperature trend in Singapore in 2013.
1. ***temp increased from Jan 2013 to May 2013 then started to decrease till Dec 2013;;***
 2. ***from 26.8°C in Jan, it increased to highest in May at 28.7°C then it decreased back to 26.8°C in Dec;;***
- [2]
- (iii) With reference to both Fig 4.2 and 4.3, account for the relationship between temperature and dengue cases in Singapore in 2013.
1. ***when temp was highest at 28.7 °C/ 28.6 °C in May/ June,***
no. of dengue cases was also highest at 842 in Week 25 (end May/ early June);;
 2. ***increase in temp increases amt of kinetic energy possessed by enz & substrates,***
thereby increasing rate of many metabolic processes;;
 3. ***shortens time taken to develop from egg to adult in Aedes aegypti,***
increases survival rate & contributes to increase in no. of Aedes aegypti which helped to spread dengue virus more rapidly;;
- [3]

- (c) Fig 4.4 is a diagram of a dengue virus.

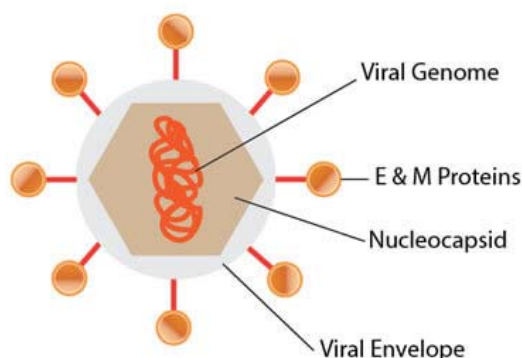


Fig. 4.4

- (i) Describe the viral genome of dengue virus.

1 single-stranded, positive-sense RNA;;

[1]

- (ii) The dengue virus and influenza virus are quite similar in terms of their structure and reproductive cycle.

Describe one structural similarity between dengue virus and influenza virus.

1. both dengue & influenza virus are enveloped virus/ surrounded by viral envelope;;

2. both possess viral glycoproteins/ proteins on their envelopes;;

[1]

- (iii) Compare the reproductive cycles of dengue virus and influenza virus.

Giving one difference and two similarities.

Difference: translation takes place immediately using host ribosomes with +ve-sense RNA genome in dengue virus

in influenza virus, synthesis of mRNA (+ve sense) have to take place first using viral RNA-dependent RNA pol before translation using host ribosome can occur;;

Similarities: 1. both enter their host cells via receptor-mediated endocytosis;;

2. acidification of endosome resulting in fusion of viral envelope with endosomal membrane occurs during uncoating in both;;

3. both viral assembly occurs at host cell's rER, where both envelope proteins are inserted into rER membrane;;

[3]

- (iv) Currently, there are no specific antiviral drugs for the treatment of dengue fever, due to the prevalence of drug resistance in dengue viruses.

Suggest one reason how drug resistance can arise in dengue viruses.

1. *dengue virus has high rate of mutation, due to lack of proofreading activity of viral RNA-dependent RNA pol, (resulting in high rate of errors during replication);; OR*
2. *genetic recombination could occur when a host cell is infected by more than one serotype, resulting in recombinant strains of virus;;*

[1]

[Total: 15]

5 Fig 5.1 and 5.2 are diagrams showing transcription and translation.

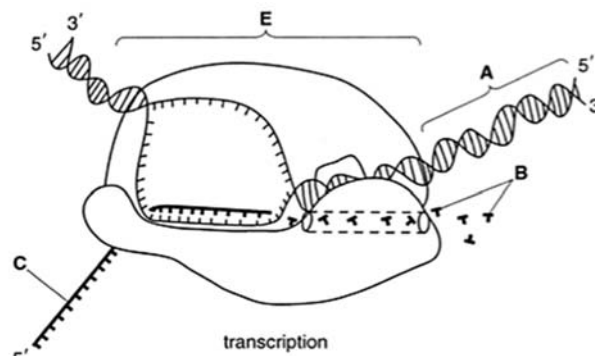


Fig 5.1

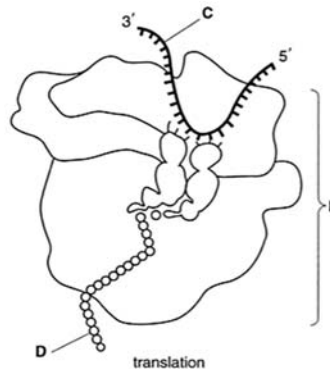


Fig 5.2

(a) Identify the structures A to C.

A **deoxyribonucleic acid** ® **abbrev. DNA**

B **ribonucleotide / ribonucleoside triphosphate** ® **nucleotide bases**

C **messenger ribonucleic acid** ® **abbrev. mRNA**

[3]

(b) Describe what happens to D after it is being synthesized to form a functional product.

1. **D will fold to form 2° struct of α -helices & β -pleated sheets**

maintained by H bonds b/w peptide bonds

2. **further coil & fold to form globular 3° struct**

maintained by hydrophobic interxns, H bonds, ionic bonds & disulfide bridges b/w R grps

[2]

(c) Describe **three** ways in which the process of transcription differs from translation.

	<i>transcription</i>	<i>translation</i>
1. <i>template</i>	<i>1 strand of DNA</i>	<i>(mature) mRNA</i>
2. <i>enz</i>	<i>RNA pol</i>	<i>peptidyl transferase in ribo</i>
3. <i>bonds</i>	<i>phosphodiester</i>	<i>peptide</i>
4. <i>monomers</i>	<i>ribonucleotides</i>	<i>aa</i>
5. <i>product</i>	<i>RNA</i>	<i>polypeptide</i>
6. <i>location</i>	<i>nucleus</i>	<i>ribosome in cytoplasm</i>
7. <i>initiation</i>	<i>TATA box of core promoter</i>	<i>start codon at 5' end</i>
8. <i>termination</i>	<i>after polyadenylation signal</i>	<i>after stop codon</i>

[3]

(d) Briefly describe how the structure of **F** differs from the structure **E**.

- F** is made up of 2 subunits, 1 large & 1 small subunit while **E** is only composed 1 entire unit / multiple subunits / enz complex
- F** consists of ribosomal RNA & ribosomal prots while **E** consist of only prot

[2]

® **comparison of binding sites**

® **comparison of initiation complexes (process shown is clearly at elongation)**

(e) Fig 5.3 shows a proteasome degrading a protein.

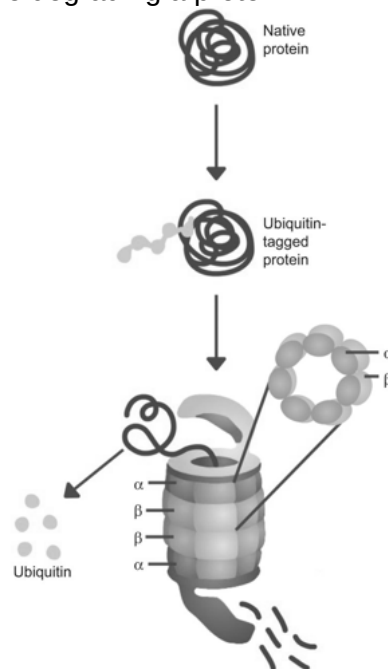


Fig. 5.3

With reference to Fig 5.3, describe what happens during proteasomal degradation of proteins.

1. **multiple ubiquitin are added to target prot by enz in cytosol**

to form ubiquitin-tagged prot

2. ***ubiquitin-tagged prot is recognised by proteasome, unfolded & enters core of proteasome***

ubiquitin is released back into cytosol

3. ***proteasome catalyse hydrolysis of peptide bond in polypep chain***

into short peptide fragments → release into cytosol for recycling

[3]

[Total: 13]

- 6 The marine threespine sticklebacks, *Gasterosteus aculeatus* is a freshwater fish living in the lakes of British Columbia, Canada as shown in Fig. 6.1.

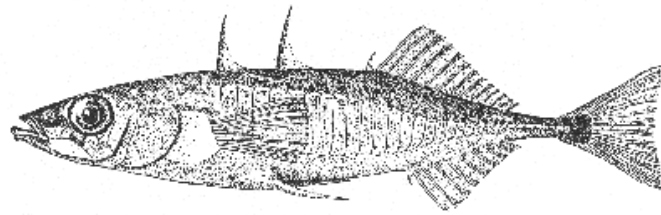


Fig. 6.1

In order to investigate the process of speciation in these populations, three small lakes were studied. Each lake contained two varieties of stickleback: a large, bottom-dwelling variety that fed on invertebrates near the shore and a small, plankton-eating variety that lived in the open water. The probability of breeding between pairs of individuals was measured under laboratory conditions in the following breeding combinations:

- I different varieties from the same lake
- II different varieties from different lakes
- III same variety from different lakes
- IV same variety from the same lake

The data are summarized in Fig. 6.2 below.

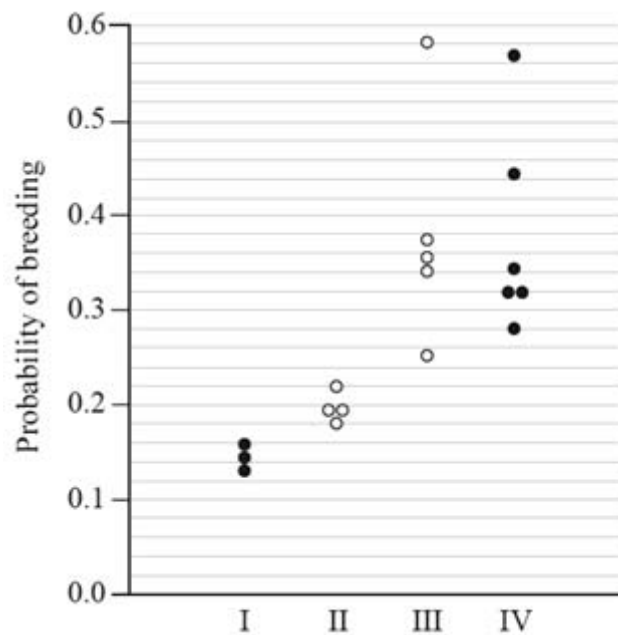


Fig 6.2

- (a) With reference to Fig 6.2,
(i) identify the highest and lowest probabilities of breeding for individuals of the same variety

highest prob is 0.58 & lowest prob is 0.25;;

[1]

- (ii) describe the differences in probability of breeding between individuals from different lake.

1. individuals from diff variety have lower breeding prob

prob b/w 0.18 – 0.22;;

2. individuals from same variety have higher breeding prob

prob b/w 0.25 - 0.58;;

[2]

- (iii) describe the evidence that speciation is taking place in these populations and explain the type of speciation

1. prob of diff varieties interbreeding is low even if they are in the same lake

probability b/w 0.13 – 0.16;;

2. sympatric speciation

behavioural or physiological isolation within same habitat;;

3. accumulation of genetic differences b/w varieties

resulting in reproductive isolation/barriers (can be pre-zygotic or post-zygotic barriers);;

[3]

- (iv) explain why all the individuals are still considered the same species.

1. based on biological species concept, both varieties can still interbreed

to produce viable, fertile offsprings;;

2. probability of breeding of different variety from same lake is 0.13-0.16

probability of breeding of different variety from different lake is 0.18-0.22;;

[2]

- (b) The freshwater lakes also contain many different types of parasites that infect the different varieties of marine threespine sticklebacks.

Explain why these parasites help to speed up speciation of the marine threespine sticklebacks.

1. diff type of parasites have specificity to infect diff varieties of animal

presence of parasites acts as a selection pressure;;

2. animals with alleles which confer resistance to parasite infection are selected for

they have high survival & reproductive rate;;

3. pass down favourable alleles to next generation

changes allelic frequency in popn of diff varieties;;

4. as genetic differences increase b/w popn of animals results in reproductive isolation

.....
prevents gene flow b/w popn & causes speciation to occur;;
.....

[4]

[Total: 12]

7 Fig. 7.1 shows how innate immune system protects the body against pathogens.

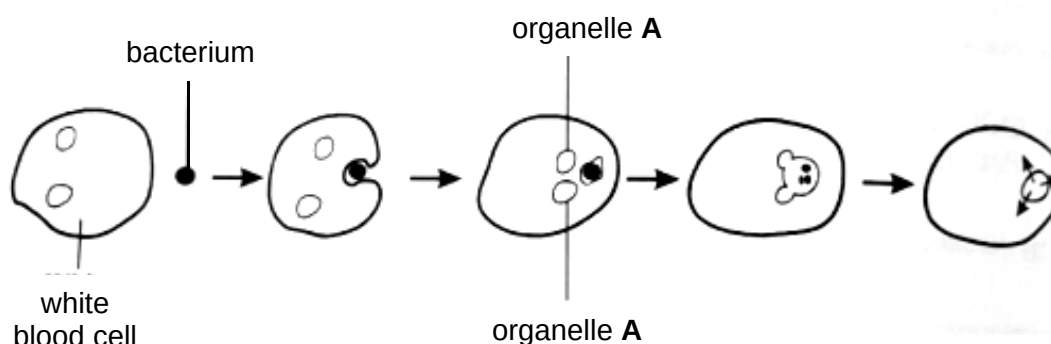


Fig. 7.1

(a) With reference to Fig 7.1,

(i) state the name of the white blood cell and organelle A.

white blood cell: macrophage/dendritic cell

organelle A: lysosome;;

[1]

(ii) describe the role of organelle A in the defence against pathogen.

1. lysosomes fuse with phagosome

to form phagolysosome/secondary lysosome;;

2. lysosome has hydrolytic enz which digests pathogen

in an acidic environment within lysosome;;

[2]

It is found that ingested *mycobacterium tuberculosis* is able to survive within the macrophage and cause tuberculosis in humans.

Bacillus Calmette–Guérin (BCG) is a vaccine primarily used against tuberculosis. It consists of live attenuated bacteria. In countries where tuberculosis is common, one dose is recommended in healthy babies as close to the time of birth. Babies with HIV/AIDS should not be vaccinated.

(b) (i) Explain how BCG vaccination provides long term immunity against tuberculosis.

1. attenuated virus retain ability to stimulate immune response

due to presence of specific surface Ags;;

2. APC take up virus by phagocytosis to present peptides of Ag

naïve helper T cell binds to Ag complex to be activated to effector T cell;;

3. effector T cell bind to Ag on naïve B cell to activate

clonal expansion of activated B cell OR clonal expansion of activated helper T cell;;

4. memory B cells & memory T cells formed via mitosis

when exposed to same pathogen, memory cells can trigger a secondary immune response;;

[4]

- (ii) Suggest why babies with HIV/AIDS should not be vaccinated
- 1. HIV/AIDS leads to weak immune system/ reduced immunity due to reduced no. of helper T cells/ B cells / action of phagocytes**
 - 2. bacteria in vaccine, can multiply faster/ are not destroyed;;**

[1]

A test as shown in Fig. 7.2 has been developed to find if a person has antibodies against *M. tuberculosis*.

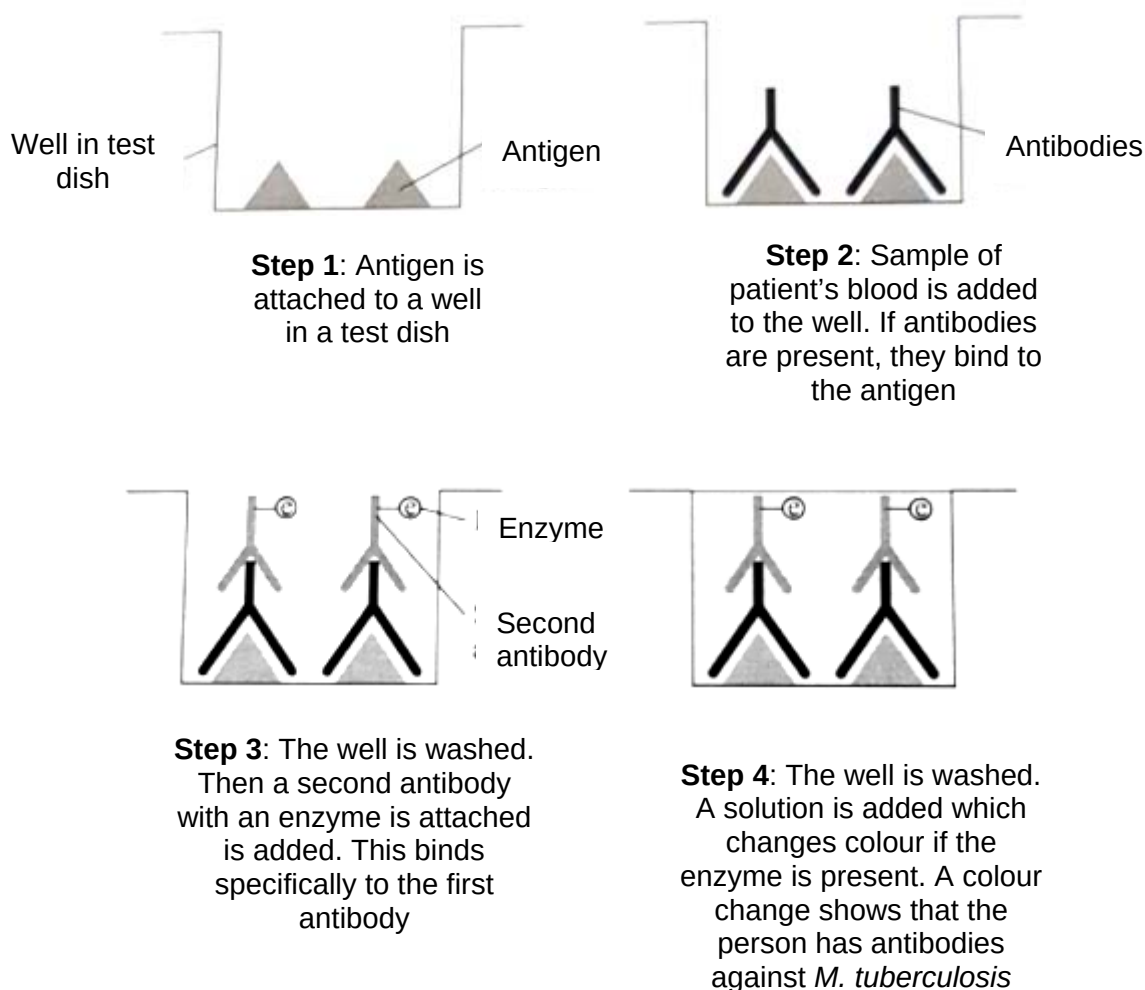


Fig. 7.2

- (c) Predict and explain if the color of the solution will change if the patient is infected with influenza virus.
- 1. No. colour of solution will not change if patient is infected with influenza virus;;**
 - 2. antigen binding site in Ab would be specific to Ag in *M. tuberculosis* will not bind to any other Ags;;**
 - 3. second Ab thus will not be able to bind to 1st Ab which would have been washed away;;**

4. thus there would be no presence of enz

colour of soln will not change without enz;;

[4]

[Total: 12]

- 8 Corals are simple marine animals and usually exist in colonies of thousands of individuals. Zooxanthellae are group of unicellular algae that can photosynthesize. They live within cells of the coral and have a symbiotic relationship.

Corals absorb calcium carbonate from the sea to build their skeletons which provides structural support. Coral reefs provide home for about 25% of known fish species.

Corals are sometimes mistaken for members of plant kingdom.

- (a) State one way in which coral cells differ from plant cells

1. coral cells do not photosynthesize/do not have chloroplasts;; OR

coral cells do not have cellulose for structural support;;

[1]

Coral reefs are at risk of damage by human activities. A study was conducted to see the effects of climate change on coral reefs.

Coral reef sites were subjected to two different environmental conditions i.e exposed site and sheltered site. Coral reefs in exposed site was exposed to climate change environmental conditions. Coral reefs in the sheltered site was exposed to normal environmental conditions.

Table 2.1 shows coral cover area at exposed and sheltered sites.

Table 2.1

Experimental site		Area of healthy coral reef/m ²	Average area of healthy coral reef/m ²
Exposed Site	Site 1	120	
	Site 2	100	
	Site 3	150	
Sheltered Site	Site 1	82	
	Site 2	75	
	Site 3	69	

- (b) With reference to Table 2.1,
(i complete Table 2.1 by calculating the average area of healthy coral reef in exposed and sheltered site.
)

$$x_1 = (120 + 100 + 150)/3 \\ = 123.3m^2$$

$$x_2 = (82 + 75 + 69)/3 \\ = 75.3m^2$$

[1]

- (ii) conduct a t -test on the given data and determine if the difference in mean area of healthy coral reef in exposed and sheltered site are statistically significant.

standard deviation $s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$

t -test $t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$ $v = n_1 + n_2 - 2$

Key to symbols

s^* = standard deviation

$\sum$ = 'sum of'

$\bar{x}$ = mean

n = sample size (number of observations) x = observation

v = degrees of freedom

t Table

cum. prob one-tail two-tails	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$
df									
1	0.000	1.000	1.378	1.963	3.078	6.314	12.71	31.82	63.66
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032
6	0.000	0.718	0.908	1.134	1.440	1.943	2.447	3.143	3.707
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169

Null hypothesis: There is no significant difference between mean area of healthy coral reef in exposed and sheltered site

$$x_1 = 123.3m^2$$

$$x_2 = 75.3m^2$$

$$s_1 = 25.166$$

$$s_2 = 6.506 ;;$$

$$n_1 = 3$$

$$n_2 = 3$$

$$t_{\text{calculated}} = 3.198 ;;$$

$$Df = 6 - 2$$

$$= 4$$

$$t_{\text{critical}} = 2.776 ;;$$

$t_{\text{calculated}}$ (3.198) is greater than t_{critical} (2.776), null hypothesis is rejected.

Conclusion: The difference in the mean area of healthy coral reef in exposed site and sheltered site is statistically significant ;; [4]

(ii) Explain two ways how climate change damages coral reefs.

1. increase in green house gasses emission in atmosphere

traps heat in atmosphere warms atmospheric temp & absorbed by water bodies/ocean;;

2. at higher water temp, increased photosynthesis rate of zooxanthellae

leading to excess product which is toxic;;

3. this damages coral causing coral polyp to expel zooxanthellae

which results in coral being bleached;;

4. ocean absorbs increased amt of carbon dioxide in the air

causes ocean acidification/drop in pH of the ocean;;

5. corals will not be able to absorb calcium carbonate

thus unable to maintain their skeleton;;

6. skeleton that provides structural support to coral dissolves

leading to death of corals;;

[6]

[Total: 12]

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*For
Examiner's
Use*